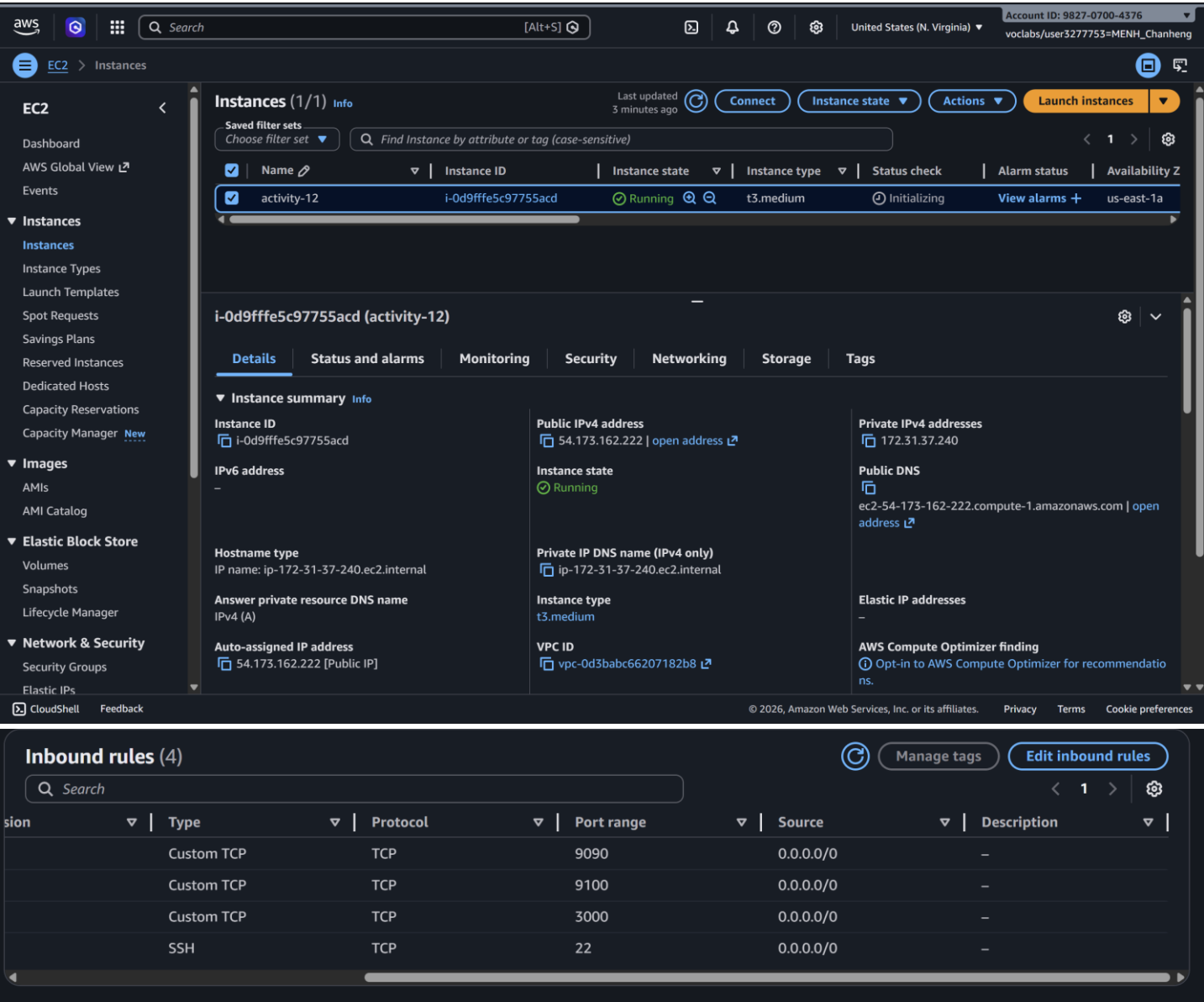


Problem 1:

EC2 Monitoring with Grafana, Prometheus, Node Exporter using Docker

Task 1: Create EC2 instance (OS – Ubuntu, Instance type - t3.medium, Create your own key-pair, Open Port Number – 22 (SSH), 3000(Grafana), 9090 (Prometheus), 9100 (Node Exporter)) - Screenshot Public IP



Task 2.1: Update OS

Task 2.2: Install and Configure Docker

Task 2.3: Install and Configure Docker Compose

Task 3.1: Create or Clone Docker Compose File from GitHub - Docker Compose Code Screenshot

Docker-compose.yml:

```
GNU nano 7.2                                docker-compose.yml *
```

```
version: '3'

services:
  prometheus:
    image: prom/prometheus
    container_name: prometheus
    volumes:
      - ./prometheus.yml:/etc/prometheus/prometheus.yml
    ports:
      - "9090:9090"

  node_exporter:
    image: prom/node-exporter
    container_name: node_exporter
    ports:
      - "9100:9100"

  grafana:
    image: grafana/grafana
    container_name: grafana
    ports:
      - "3000:3000"
```

Task 3.2: Create or Clone Prometheus and Node Export configuration File from GitHub - Code Screenshot

```
GNU nano 7.2                                prometheus.yml *
```

```
global:
  scrape_interval: 15s

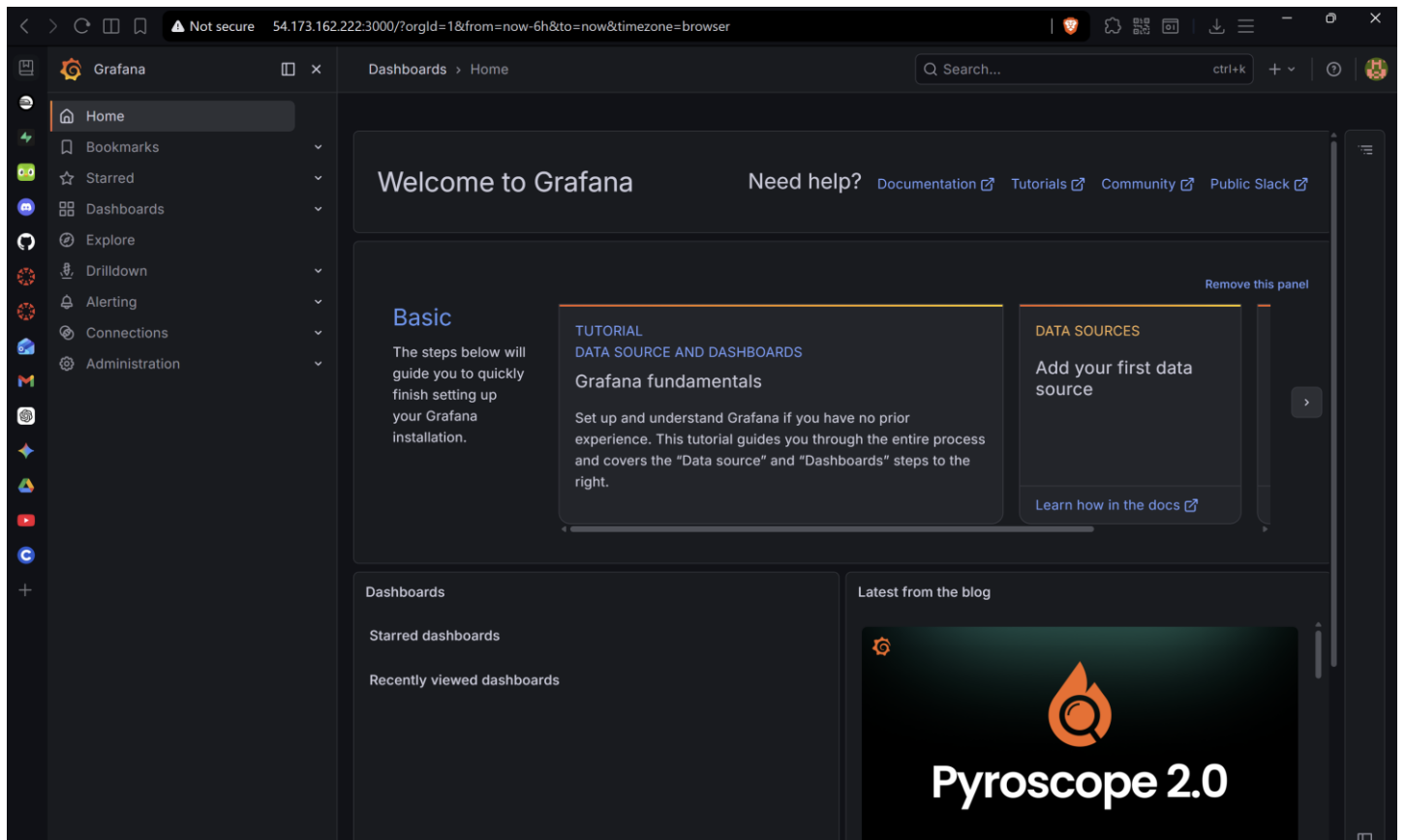
scrape_configs:
  - job_name: 'node_exporter'
    static_configs:
      - targets: ['node_exporter:9100']
```

Task 4: Run Docker compose

```
aws  [Search] [Alt+S]  United States (N. Virginia)  Account ID: 9827-0700-4376
voclabs/user3277753=MENH_Chanheng

ubuntu@ip-172-31-37-240:~/monitoring$ docker-compose up -d
Creating network "monitoring default" with the default driver
Pulling prometheus (prom/prometheus):...
latest: Pulling from prom/prometheus
a6680b927350: Pull complete
f60c291aa4d9: Pull complete
a2243411ec8e: Pull complete
ce2ce3cfbbe4: Pull complete
e3e60763add3: Pull complete
5b801931d496: Pull complete
9fa05948455e: Pull complete
d70e68dbd61c: Pull complete
f86ef4240a84: Pull complete
259391b25b59: Pull complete
Digest: sha256:5550dc63da361dc30f6fe02ac0e4dfc736ededef3c8d12a634db04a67824d78
Status: Downloaded newer image for prom/prometheus:latest
Pulling node_exporter (prom/node-exporter):...
latest: Pulling from prom/node-exporter
fe82e83000f4: Pull complete
Digest: sha256:e9cff4fc67b1818f8c97adb115b9f12c9a54b533de86765d4a0effc01b357205
Status: Downloaded newer image for prom/node-exporter:latest
Pulling grafana (grafana/grafana):...
latest: Pulling from grafana/grafana
0083d589f79f: Pull complete
5f4c407a6406: Pull complete
3616d2fa29d1: Pull complete
fa48d0c72410: Pull complete
589002ba0eae: Pull complete
5b4a39de381d: Pull complete
a154139e28ac: Pull complete
bcb2a7709afa: Pull complete
9f73c35df60c: Pull complete
d2400e864d89: Pull complete
d62f64b7b42b: Pull complete
4f4fb700ef54: Pull complete
08483b2bdbb1: Pull complete
Digest: sha256:0f86bada30d65ef9d0183b90c1e2682ac92d53d95da8bed322b984ea78a4a73a
Status: Downloaded newer image for grafana/grafana:latest
Creating prometheus ... done
Creating node_exporter ... done
Creating grafana ... done
ubuntu@ip-172-31-37-240:~/monitoring$
```

Task 5: Open Grafana and Login - Grafana After Login Screenshot



Task 6: Add Prometheus Data Source and Save, Test It - Test Success Screenshot

Home

Bookmarks

Starred

Dashboards

Explore

Drilldown

Alerting

Connections

Add new connection

Data sources

Administration

Connections > Data sources > prometheus-4

Search...

ctrl+k

prometheus-4

Type: Prometheus

Alerting

Supported

Explore data

Build a dashboard

Settings

Permissions

Insights

Cache

Configure your Prometheus data source below

Or skip the effort and get Prometheus as fully-managed, scalable, and hosted data source from Grafana Labs with the [free-forever Grafana Cloud plan](#).

Name

prometheus-4

Default

Before you can use the Prometheus data source, you must configure it below or in the config file. For detailed instructions, [view the documentation](#).

Fields marked with * are required

Connection

Prometheus server URL *

http://prometheus:9090

Authentication

Authentication methods

Custom query parameters

Example: max_source_resolution=5m&in

HTTP method

POST

Series limit

40000

Use series endpoint

Exemplars

+ Add

Successfully queried the Prometheus API.

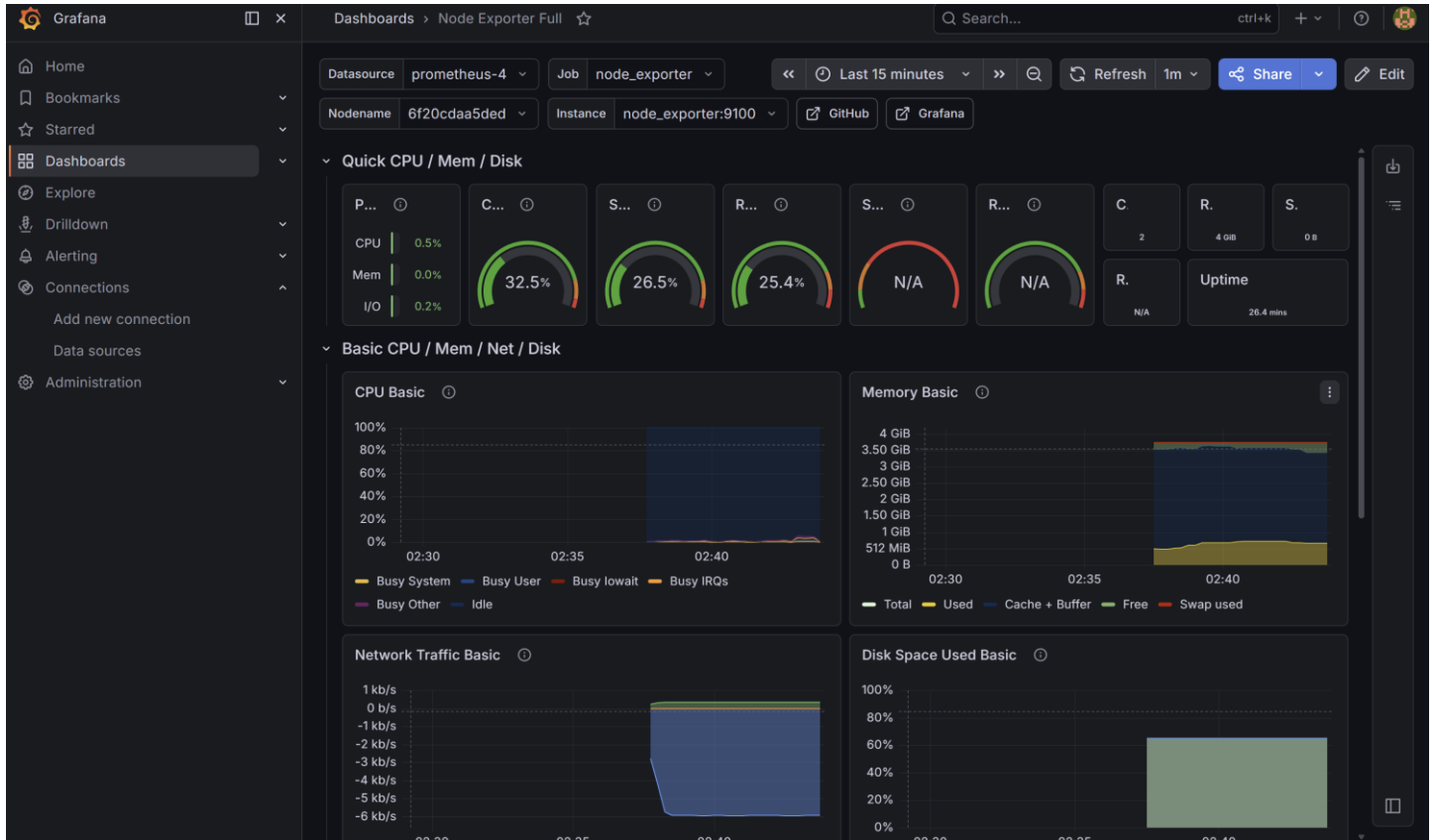
Next, you can start to visualize data by [building a dashboard from scratch](#) or by querying data in the Explore view.

Open in Metrics Drilldown

Delete

Save & test

Task 7: Import and View Dashboard (id - 1860) - Dashboard Screenshot



Task 8: Install stress and start it

```
ubuntu@ip-172-31-37-240:~/monitoring$ sudo apt install -y stress
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  stress
0 upgraded, 1 newly installed, 0 to remove and 64 not upgraded.
Need to get 18.1 kB of archives.
After this operation, 52.2 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 stress amd64 1.0.7-1 [18.1 kB]
Fetched 18.1 kB in 0s (1052 kB/s)
Selecting previously unselected package stress.
(Reading database ... 73076 files and directories currently installed.)
Preparing to unpack .../stress_1.0.7-1_amd64.deb ...
Unpacking stress (1.0.7-1) ...
Setting up stress (1.0.7-1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

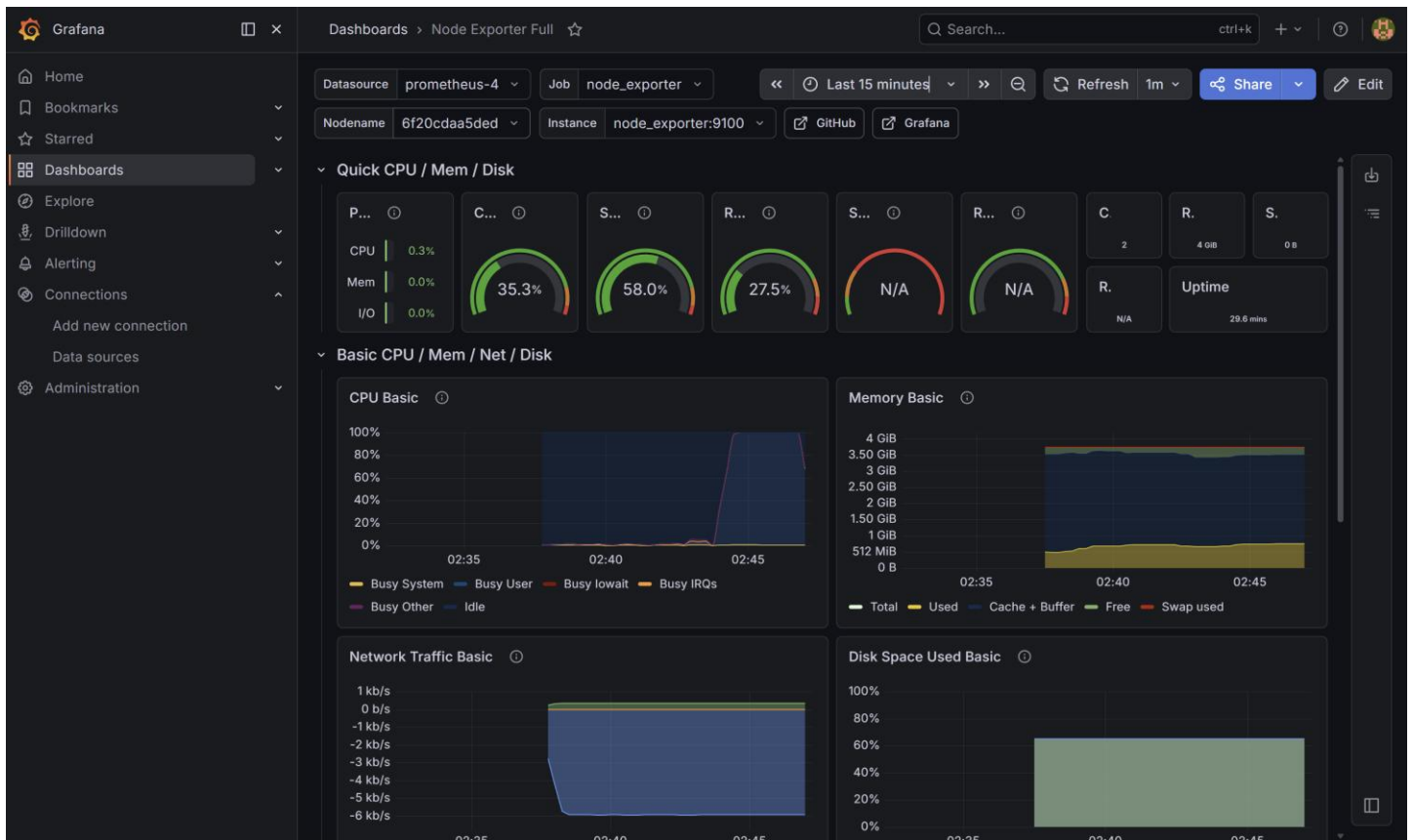
No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-37-240:~/monitoring$ stress --cpu 2 --timeout 180
stress: info: [4027] dispatching hogs: 2 cpu, 0 io, 0 vm, 0 hdd
stress: info: [4027] successful run completed in 180s
ubuntu@ip-172-31-37-240:~/monitoring$
```

Task 9: View Dashboard (id - 1860) - Dashboard Screenshot



Question: Do you find any changes in your Dashboard Graph?

Answer: Yes

- CPU usage increased significantly during stress execution
- Load average also increased
- This confirms:
 - Node Exporter collects system metrics
 - Prometheus scrapes them
 - Grafana visualizes real-time changes

Problem 2:

Nginx or any other Server Monitoring with Grafana, Prometheus, Node Exporter using Docker or Manual Installation of Grafana, Prometheus, Node Exporter etc.

If you use docker screenshot only full code docker compose, Prometheus, node export configuration file and final dashboard Screenshot

If you use manual installation all configuration file screenshot and final dashboard Screenshot

nginx.conf:

```
GNU nano 7.2 nginx.conf *
events {}

http {
    server {
        listen 80;

        location / {
            return 200 "Nginx is running\n";
        }

        location /stub_status {
            stub_status;
            allow all;
        }
    }
}
```

docker-compose.yml:

```
GNU nano 7.2 docker-compose.yml *
version: '3'

services:
  prometheus:
    image: prom/prometheus
    container_name: prometheus
    volumes:
      - ./prometheus.yml:/etc/prometheus/prometheus.yml
    ports:
      - "9090:9090"

  node_exporter:
    image: prom/node-exporter
    container_name: node_exporter
    ports:
      - "9100:9100"

  grafana:
    image: grafana/grafana
    container_name: grafana
    ports:
      - "3000:3000"

  nginx:
    image: nginx:latest
    container_name: nginx
    volumes:
      - ./nginx.conf:/etc/nginx/nginx.conf:ro
    ports:
      - "80:80"

  nginx_exporter:
    image: nginx/nginx-prometheus-exporter
    container_name: nginx_exporter
    command:
      - '-nginx.scrape-uri=http://nginx/stub_status'
    ports:
      - "9113:9113"
```

prometheus.yml:

```
GNU nano 7.2 prometheus.yml *
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: 'node_exporter'
    static_configs:
      - targets: ['node_exporter:9100']

  - job_name: 'nginx'
    static_configs:
      - targets: ['nginx_exporter:9113']

ubuntu@ip-172-31-37-240:~/monitoring$ docker-compose down
Stopping grafana      ... done
Stopping prometheus   ... done
Stopping node_exporter ... done
Removing grafana      ... done
Removing prometheus   ... done
Removing node_exporter ... done
Removing network monitoring_default
ubuntu@ip-172-31-37-240:~/monitoring$ docker-compose up -d
Creating network "monitoring_default" with the default driver
Pulling nginx (nginx:latest)...
latest: Pulling from library/nginx
801a1ad15b4e: Pull complete
85c66128325a: Pull complete
3531af2bc2a9: Pull complete
ce776bbcdad0d: Pull complete
ff048f1f2159: Pull complete
4677c2a9a3d4: Pull complete
677c63196868: Pull complete
69989ccd189b: Download complete
54c38c75806e: Download complete
Digest: sha256:6e23479198b998e5e25921dff8455837c7636a67111a04a635cf1bb363d199dc
Status: Downloaded newer image for nginx:latest
Pulling nginx_exporter (nginx/nginx-prometheus-exporter)...
latest: Pulling from nginx/nginx-prometheus-exporter
49619ac15983: Pull complete
13b555b32501: Pull complete
5be3366d8afd: Download complete
d2f7fc3af37c: Download complete
Digest: sha256:bf76e58df548a13e52bd543ac70f6d3d667e14884e8b51c149bd81994ca75ccd
Status: Downloaded newer image for nginx/nginx-prometheus-exporter:latest
Creating node_exporter ... done
Creating nginx_exporter ... done
Creating grafana      ... done
Creating nginx        ... done
Creating prometheus   ... done
ubuntu@ip-172-31-37-240:~/monitoring$

ubuntu@ip-172-31-37-240:~/monitoring$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
d3f719b173f0   nginx:latest                        "/docker-entrypoint..." 22 seconds ago Up 21 seconds 0.0.0.0:80->80/tcp, [::]:80->80/tcp
nginx
49c10666f7e7   nginx/nginx-prometheus-exporter     "/usr/bin/nginx-prom..." 22 seconds ago Up 21 seconds 0.0.0.0:9113->9113/tcp, [::]:9113->9113/tcp
nginx_exporter
bab8e8e404daa   prom/prometheus                    "/bin/prometheus --c..." 22 seconds ago Up 21 seconds 0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp
prometheus
d9d08ad26e75    grafana/grafana                    "/run.sh"                 22 seconds ago Up 21 seconds 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
grafana
5e1bbca9803c    prom/node-exporter                 "/bin/node_exporter"      22 seconds ago Up 21 seconds 0.0.0.0:9100->9100/tcp, [::]:9100->9100/tcp
node_exporter
ubuntu@ip-172-31-37-240:~/monitoring$
```

Validate nginx:

```
ubuntu@ip-172-31-37-240:~/monitoring$ curl http://localhost
Nginx is running
ubuntu@ip-172-31-37-240:~/monitoring$ curl http://localhost/stub_status
Active connections: 2
server accepts handled requests
 3 3 9
Reading: 0 Writing: 1 Waiting: 1
ubuntu@ip-172-31-37-240:~/monitoring$

ubuntu@ip-172-31-37-240:~/monitoring$ curl http://localhost:9113/metrics
# HELP go_gc_duration_seconds A summary of the wall-time pause (stop-the-world) duration in garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 0.000155925
go_gc_duration_seconds{quantile="0.25"} 0.000155925
go_gc_duration_seconds{quantile="0.5"} 0.000155925
go_gc_duration_seconds{quantile="0.75"} 0.000155925
go_gc_duration_seconds{quantile="1"} 0.000155925
go_gc_duration_seconds_sum 0.000155925
go_gc_duration_seconds_count 1
# HELP go_gc_gogc_percent Heap size target percentage configured by the user, otherwise 100. This value is set by the GOGC environment variable, and the runtime/debug.SetGCPercent function. Sourced from /gc/gogc:percent.
# TYPE go_gc_gogc_percent gauge
go_gc_gogc_percent 100
# HELP go_gc_gomemlimit_bytes Go runtime memory limit configured by the user, otherwise math.MaxInt64. This value is set by the GOMEMLIMIT environment variable, and the runtime/debug.SetMemoryLimit function. Sourced from /gc/gomemlimit:bytes.
# TYPE go_gc_gomemlimit_bytes gauge
go_gc_gomemlimit_bytes 9.223372036854776e+18
# HELP go_goroutines Number of goroutines that currently exist.
# TYPE go_goroutines gauge
go_goroutines 14
# HELP go_info Information about the Go environment.
# TYPE go_info gauge
go_info{version="go1.25.1"} 1
# HELP go_memstats_alloc_bytes Number of bytes allocated in heap and currently in use. Equals to /memory/classes/heap/objects:bytes.
# TYPE go_memstats_alloc_bytes gauge
go_memstats_alloc_bytes 3.146608e+06
# HELP go_memstats_alloc_bytes_total Total number of bytes allocated in heap until now, even if released already. Equals to /gc/heap/allocs:bytes.
# TYPE go_memstats_alloc_bytes_total counter
```


Prometheus

QueryAlertsStatus > Target health

1234

Select scrape pool

Filter by target health

Filter by endpoint or labels

nginx

1 / 1 up

Endpoint	Labels	Last scrape	State
http://nginx_exporter:9113/metrics	instance="nginx_exporter:9113"job="nginx"	7.969s ago2ms	UP

node_exporter

1 / 1 up

Endpoint	Labels	Last scrape	State
http://node_exporter:9100/metrics	instance="node_exporter:9100"job="node_exporter"	15.051s ago26ms	UP

Grafana

Dashboards > Import dashboard

Search...

Home

Bookmarks

Starred

Dashboards

Explore

Drilldown

Alerting

Connections

Add new connection

Data sources

Administration

Import dashboard

Import dashboard from file or Grafana.com

Importing dashboard from Grafana.com

Published by

alvarobacelar

Updated on

2020-07-26 09:23:16

Options

Name

NGINX exporter

Folder

Dashboards

Unique identifier (UID)

The unique identifier (UID) of a dashboard can be used to uniquely identify a dashboard between multiple Grafana installs. The UID allows having consistent URLs for accessing dashboards so changing the title of a dashboard will not break any bookmarked links to that dashboard.

MsjffzSZz

Change uid

DS_PROMETHEUS

Select a prometheus data source

prometheus

Import

Cancel

Grafana | Dashboards > NGINX exporter

datasource: default | instance: All x x

« | Last 15 minutes | » | 🔍 | 🔄 Refresh 5s | ➦ Share | ✎ Edit

Status

NGINX Status for nginx_exporter:9113

Up

Metrics

Processed connections

Connections (rate)

0.1
0.1
0.1
0.0
0.0
0.0
0.0
0.0

02:55 03:00 03:05

— nginx_exporter:9113 accepted — nginx_exporter:9113 handled

Active Connections

Connections

2
1
0

02:55 03:00 03:05

— nginx_exporter:9113 active — nginx_exporter:9113 reading
— nginx_exporter:9113 waiting — nginx_exporter:9113 writing

Total requests

0.14
0.12
0.1

02:55 03:00 03:05

— nginx_exporter:9113 accepted — nginx_exporter:9113 handled